

# Reciprocal-Integer Convolution Divisors of the Rvachev Law

Arithmetic semigroups, singular-to-smooth parity, Thue–Morse quotients, zero zeta functions, and inverse-Fabius approximants

Research report prepared from the ProveIt Fabius-function corpus

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## Abstract

Let  $X$  be the compactly supported random variable whose density is the Rvachev up-function and whose characteristic function, in the Fourier convention used here, is

$$\Phi(z) = \prod_{n \geq 0} \operatorname{sinc}_{\text{rad}}\left(\frac{\pi z}{2^n}\right).$$

This report develops a new arithmetic family of exact convolution divisors of the Rvachev law. For every integer  $M \geq 2$  the quotient

$$Q_M(z) = \frac{\Phi(z)}{\Phi(z/M)}$$

extends through all apparent zero-over-zero points and is a characteristic function. Its law  $\mathbf{R}_M^{\text{law}}$  satisfies

$$X \stackrel{d}{=} \frac{X'}{M} + Y_M, \quad Y_M \sim \mathbf{R}_M^{\text{law}}, \quad X' \perp Y_M.$$

Conversely, these and their reflected versions are the only nontrivial independent-copy scale decompositions of  $X$ : if  $X \stackrel{d}{=} cX' + Y$ , then  $c = 0$  or  $c = \pm 1/M$  for an integer  $M \geq 1$ .

The divisors form a multiplicative cocycle,  $Q_{MN}(z) = Q_M(z)Q_N(z/M)$ , and admit an elementary digit model. After an affine change of variables,  $\mathbf{R}_M^{\text{law}}$  is generated by i.i.d. digits uniform on  $\{0, \dots, M-1\}$  in a binary radix. This produces a complete parity and regularity classification. Odd  $M$  give singular continuous full-support measures with a nondecaying Fourier subsequence. If  $M = 2^r q$  with  $q$  odd, then even  $M$  are absolutely continuous; for  $q > 1$  their densities are exactly  $C^{r-1}$  but not  $C^r$ , while  $q = 1$  gives finite unequal-knot B-splines. Thus canonical singular measures with interval support approach the  $C^\infty$  Rvachev law at the exact  $W_\infty$  rate  $1/M$ .

Several further structures follow. The zero divisor of  $Q_M$  has an explicit 2-adic multiplicity formula and spectral zeta function

$$\mathcal{Z}_M(s) = (1 - M^{-s}) \frac{\zeta(s)}{1 - 2^{-s}}.$$

Its dyadic Euler cancellation is complete precisely for powers of two. The finite digit polynomials satisfy the exact Thue–Morse quotient identity

$$A_{M,N}(x)T_N(x) = T_N(x^M),$$

and for  $M = 3$  their stabilized coefficients are the hyperbinary counts  $s(n+1)$  from Stern's diatomic sequence. Cumulants are explicit Bernoulli-number multiples and moments are complete Bell polynomials. Endpoint distribution functions possess exact log-periodic scaling; the phase is constant if and only if  $M$  is a power of two. Finally, the decomposition yields certified uniform

approximants to the inverse Fabius function, a radix- $M$  innovation expansion, and a natural  $q = M^{-2}$  Richardson scheme. The report closes with conjectures on fractional regularity, complex-dimension expansions, dyadic arithmetic, Legendre transfer operators, general bases, and formalization.

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**Status and originality convention.** Every statement labeled theorem, proposition, lemma, corollary, or identity is proved in this report, except when explicitly attributed to a cited source. Every conjecture is visibly labeled. “New” or “frontier” means that the result was not found in the audited 30 August 2026 repository corpus or in the limited external literature search described in [section 1](#); it is not a claim of mathematical priority. The arguments have not been peer reviewed or formally checked in Lean.

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# 1 Corpus audit, scope, and novelty controls

## 1.1 Operational reading protocol

The requested corpus contains current expositions, large consolidated frontier volumes, small living drafts, and a frozen archive with many absorbed or near-duplicate historical versions. Treating every archived snapshot as an independent paper would inflate apparent coverage and obscure the active novelty baseline. The audit therefore used four layers:

1. every visible `.tex` path below `Analysis/FabiusFunction/docs` was inventoried through the repository tree, frontier manifest, and archive provenance lists;
2. the living primary exposition and glossary were read for definitions, normalizations, proved Fourier products, zero multiplicities, arithmetic values, and asymptotic conventions;
3. the current consolidated frontier volumes were read through their abstracts, tables of contents, theorem summaries, novelty audits, and targeted full-text searches;
4. frozen historical files were used as a provenance and duplication map, with the active consolidations treated as the mathematical source of record.

This is more reliable for nonduplication than counting hundreds of archived revisions as separate works. It is also transparent about what “read all documents” means in a large, versioned research corpus: every path and provenance record was inspected, while the living and consolidated sources received the substantive mathematical reading.

The principal active sources were the current expositions [1, 2]

- `Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`,
- `FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex`,
- `Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex`,
- `fabius_lean_walkthrough/fabius_lean_walkthrough.tex`,
- `semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex`,

together with the current consolidated volumes on Thue–Morse identities, exponents and  $q$ -series, inverse and sampling theory, representation theory, and frontier compilations. The repository already contains extensive work on endpoint Lambert- $W$  asymptotics, Bell–Bernoulli expansions,  $q$ -binomial extrapolation, integer-zero filtrations, spectral zeta functions attached to the original  $\Phi$ , Legendre and Lagrange loops, finite uniform splines, multiresolution, and general atomic functions. Those themes are used here as infrastructure rather than repeated as purportedly new results.

## 1.2 Targeted duplication tests

The active full-text corpus was searched for the phrases and formula patterns

|                       |                        |                            |                           |
|-----------------------|------------------------|----------------------------|---------------------------|
| self-decomposable     | reciprocal integer     | convolution divisor        | Dirichlet kernel quotient |
| integer scale divisor | multiplicative cocycle | hyperbinary                | Stern                     |
| $T_N(x^M)/T_N(x)$     | singular divisor       | $Q_M = \Phi/\Phi(\cdot/M)$ | torus aliasing            |

No matching development of the family studied below was found in the living primary exposition, the Thue–Morse atlas, the exponents/ $q$ -series volume, the inverse/sampling volume, the representation volume, or the frontier compilation. The archive manifests likewise did not identify an absorbed

draft with this focus. In particular, the repository’s known dyadic truncation identity is not the complete reciprocal-integer classification of [theorem 3.3](#), and its finite uniform splines are only the power-of-two branch of the much larger family  $\{R_M^{\text{law}}\}$ .

The external search used the standard Fabius/Rvachev literature [5, 3, 4], Arias de Reyna’s arithmetic work, two-scale refinement theory [7, 8], the Jessen–Wintner pure-type principle [6], modern surveys on self-similar measures [10], and the literature on hyperbinary expansions and Stern polynomials [9]. No source found in that search treated the specific quotient  $\Phi(z)/\Phi(z/M)$  as a complete arithmetic divisor family of the Rvachev law. Some individual ingredients—Dirichlet-kernel identities, pure-type arguments, and Stern’s hyperbinary interpretation—are classical; the frontier contribution is their exact assembly around the Rvachev sinc product.

### 1.3 Map of genuinely distinct results

The report’s main additions relative to the audited corpus are:

1. the exact construction and complete classification of all independent-copy scaling divisors of the Rvachev law;
2. the multiplicative convolution cocycle  $R_{MN}^{\text{law}} = R_M^{\text{law}} * \mathcal{D}_{1/M} R_N^{\text{law}}$ ;
3. a complete odd/even and 2-adic regularity classification of  $R_M^{\text{law}}$ ;
4. singular continuous, full-interval approximants converging to up at exact  $W_\infty$  distance  $1/M$ ;
5. the zero multiplicity formula, zero zeta function, and power-of-two complex-pole rigidity for  $Q_M$ ;
6. the Thue–Morse quotient identity for bounded binary-partition polynomials and its  $M = 3$  Stern/hyperbinary specialization;
7. exact endpoint log-periodic CDF and quantile laws, with constant phase if and only if  $M$  is a power of two;
8. certified inverse-Fabius quantile brackets and radix- $M$  innovation approximants;
9. Bernoulli cumulants, Bell moments, Appell cocycles, and a finite Legendre transfer formula specialized to the divisor family.

## 2 The Rvachev probability law and its established Fourier backbone

### 2.1 Normalization

Throughout,

$$\text{sinc}_{\text{rad}} u = \begin{cases} \sin u/u, & u \neq 0, \\ 1, & u = 0, \end{cases} \quad \widehat{\mu}_{2\pi}(-z) = \int_{\mathbb{R}} e^{2\pi izx} \mu(dx).$$

Let  $X$  be the random variable with density  $\text{up}(x)$ , supported on  $[-1, 1]$ . The repository’s primary exposition records the entire characteristic function [1]

$$\Phi(z) = \widehat{\text{Law}(X)}_{2\pi}(-z) = \prod_{n=0}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{\pi z}{2^n}\right), \quad z \in \mathbb{C}. \quad (1)$$

Equivalently, if  $U_0, U_1, \dots$  are independent and uniform on  $[-1, 1]$ , then

$$X \stackrel{d}{=} \sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}}. \quad (2)$$

The convergence is absolute almost surely, the support is  $[-1, 1]$ , and

$$\mathbb{E}X = 0, \quad \text{Var}(X) = \frac{1}{9}. \quad (3)$$

Let  $F_{\text{bd}}$  be the standard Fabius function on  $[0, 1]$ . In this normalization,

$$\mathbb{P}(X \leq t) = F_{\text{bd}}\left(\frac{t+1}{2}\right), \quad -1 \leq t \leq 1, \quad (4)$$

and  $F'_{\text{bd}}(x) = 2 \text{up}(2x-1)$ . This distinction is important:  $F_{\text{bd}}$  is the affine CDF of the up density, while the identity  $F_{\text{bd}}(x) = \text{up}(x-1)$  on  $[0, 1]$  is a special functional coincidence of the Fabius/Rvachev pair.

## 2.2 The integer zero lattice

A second established fact used repeatedly below is the complete zero divisor of  $\Phi$  [1]. For  $m \in \mathbb{Z} \setminus \{0\}$ ,

$$\text{ord}_{z=m} \Phi(z) = 1 + \nu_2(|m|), \quad (5)$$

where  $\nu_2(m)$  is the exponent of 2 in  $m$ . There are no other zeros. The canonical product can therefore be written

$$\Phi(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{1+\nu_2(n)}. \quad (6)$$

The convergence is locally uniform because  $\sum_{n \geq 1} (1 + \nu_2(n))/n^2 < \infty$ .

These two pieces—the random uniform series and the exact integer zero lattice—are the only deep repository inputs required for the main classification theorem. Everything else is derived below.

## 3 Reciprocal-integer convolution divisors

### 3.1 A Dirichlet-kernel factor at every dyadic level

Fix an integer  $M \geq 2$  and define initially away from removable singularities

$$Q_M(z) = \frac{\Phi(z)}{\Phi(z/M)}. \quad (7)$$

For  $y \in \mathbb{C}$  the elementary identity

$$\frac{\text{sinc}_{\text{rad}}(My)}{\text{sinc}_{\text{rad}}(y)} = \frac{\sin(My)}{M \sin y} = \frac{1}{M} \sum_{j=0}^{M-1} e^{i(M-1-2j)y} \quad (8)$$

holds by analytic continuation at the common zeros. Applying it with  $y = \pi z/(M2^n)$  gives

$$\frac{\text{sinc}_{\text{rad}}(\pi z/2^n)}{\text{sinc}_{\text{rad}}(\pi z/(M2^n))} = \frac{1}{M} \sum_{j=0}^{M-1} \exp\left(\frac{\pi i z(M-1-2j)}{M2^n}\right). \quad (9)$$

Let  $D_M$  be uniform on the symmetric  $M$ -point set

$$\left\{ \frac{M-1-2j}{M} : 0 \leq j < M \right\}. \quad (10)$$

Then the right side of (9) is the characteristic function of  $D_M/2^{n+1}$ . The factors equal  $1 + \mathcal{O}_K(4^{-n})$  uniformly for  $z$  in a compact set  $K \subset \mathbb{C}$ , so their product converges locally uniformly.

**Theorem 3.1** (Reciprocal-integer divisor theorem). *For every integer  $M \geq 2$ ,  $Q_M$  extends to an even entire characteristic function, with product representation*

$$Q_M(z) = \prod_{n=0}^{\infty} \frac{\text{sinc}_{\text{rad}}(\pi z/2^n)}{\text{sinc}_{\text{rad}}(\pi z/(M2^n))}. \quad (11)$$

If  $D_{M,0}, D_{M,1}, \dots$  are independent copies of  $D_M$ , then

$$Y_M := \sum_{n=0}^{\infty} \frac{D_{M,n}}{2^{n+1}} \quad (12)$$

has characteristic function  $Q_M$ , support

$$\text{supp}(R_M^{\text{law}}) = \left[ -\frac{M-1}{M}, \frac{M-1}{M} \right], \quad R_M^{\text{law}} := \text{Law}(Y_M), \quad (13)$$

and satisfies the exact independent-copy decomposition

$$X \stackrel{d}{=} \frac{X'}{M} + Y_M, \quad X' \perp Y_M, \quad X' \stackrel{d}{=} X. \quad (14)$$

*Proof.* The local uniform convergence just noted makes (11) entire. Each finite product is the characteristic function of a finite sum of independent bounded digits, and the series (12) converges absolutely. Hence the limit is the characteristic function of  $Y_M$ . The support bound follows from

$$\sum_{n \geq 0} \frac{|D_{M,n}|}{2^{n+1}} \leq \frac{M-1}{M}.$$

Both endpoints belong to the support by choosing all extreme digits, and the support is an interval by the affine model in section 4. Finally,

$$\Phi(z/M)Q_M(z) = \Phi(z),$$

so uniqueness of characteristic functions gives (14).  $\square$

*Remark 3.2.* For  $M = 2$ ,  $D_2$  is uniform on  $\{-1/2, 1/2\}$  after the first scaling in (12), and  $Y_2$  is uniform on  $[-1/2, 1/2]$ . This recovers the familiar first dyadic factor. The point of theorem 3.1 is that every integer  $M$ , not only a power of two, produces an exact independent divisor.

### 3.2 Complete classification of independent-copy scales

**Theorem 3.3** (Complete reciprocal-scale classification). *Let  $c \in \mathbb{R}$ . There exist independent random variables  $X' \stackrel{d}{=} X$  and  $Y$  such that*

$$X \stackrel{d}{=} cX' + Y \quad (15)$$

if and only if

$$c = 0 \quad \text{or} \quad c = \pm \frac{1}{M} \quad \text{for some integer } M \geq 1. \quad (16)$$

For  $M \geq 2$  the nondegenerate remainder is  $R_M^{\text{law}}$  up to reflection, while  $M = 1$  gives the degenerate remainder 0.

*Proof.* The cases in (16) exist by theorem 3.1, symmetry of  $X$ , and the trivial  $c = 0$  or  $|c| = 1$  decompositions.

Conversely, realize  $W = cX' + Y$  with  $W \stackrel{d}{=} X$ . Since  $W$  and  $X'$  are bounded,  $Y = W - cX'$  is bounded. Its characteristic function  $\psi$  is therefore entire, and

$$\Phi(z) = \Phi(cz)\psi(z) \quad (17)$$

holds first for real  $z$  and then on all of  $\mathbb{C}$  by the identity theorem. If  $c \neq 0$ , substitute  $z = 1/c$ . The factor  $\Phi(cz) = \Phi(1)$  vanishes, so  $\Phi(1/c) = 0$ . The zero classification (5) forces  $1/c$  to be a nonzero integer. Thus  $c = \pm 1/M$ .  $\square$

*Remark 3.4* (Semiselfdecomposability, but only arithmetically). In probability terminology, [theorem 3.3](#) says that the Rvachev law is  $c$ -decomposable exactly on the closed reciprocal-integer set  $\{0\} \cup \{\pm 1/M : M \in \mathbb{N}_0\}$ . It is not self-decomposable in the classical sense, which would require every  $0 < c < 1$  and would in particular force a zero-free characteristic function.

### 3.3 New refinement equations for $\text{up}$ and $F_{\text{bd}}$

Write  $\mathcal{D}_a\mu$  for the pushforward of a measure under  $x \mapsto ax$ . Then (14) is the measure identity

$$\mu = \mathcal{D}_{1/M}\mu * \mathbf{R}_M^{\text{law}}, \quad \mu(dx) = \text{up}(x) dx. \quad (18)$$

Because  $\mu$  has a smooth density, convolution gives a family of exact non-dyadic integral refinements.

**Corollary 3.5** (Integral refinement family). *For every integer  $M \geq 2$ ,*

$$\text{up}(x) = M \int_{\mathbb{R}} \text{up}(M(x - y)) \mathbf{R}_M^{\text{law}}(dy). \quad (19)$$

*If  $F_{\text{bd}}$  is extended by 0 on  $(-\infty, 0]$  and by 1 on  $[1, \infty)$ , then*

$$F_{\text{bd}}(u) = \int_{\mathbb{R}} F_{\text{bd}}\left(\frac{M(2u - 1 - y) + 1}{2}\right) \mathbf{R}_M^{\text{law}}(dy). \quad (20)$$

*Proof.* The first formula is the density of (18). For the second, apply the CDF identity to  $G(t) = F_{\text{bd}}((t + 1)/2)$ :  $G(t) = \int G(M(t - y)) \mathbf{R}_M^{\text{law}}(dy)$ , then set  $t = 2u - 1$ .  $\square$

The operator in (20) is not a rephrasing of the standard dyadic functional-differential equation. Its kernel changes arithmetic type with  $M$ : singular for odd  $M$ , finitely smooth for even non-powers of two, and a compact spline for powers of two. This variability is the source of the next sections.

## 4 The digit IFS and the multiplicative convolution cocycle

### 4.1 Affine digit coordinates

Put

$$Z_M := \sum_{n=1}^{\infty} \frac{J_n}{2^n}, \quad J_n \stackrel{\text{iid}}{\sim} \text{Unif}\{0, 1, \dots, M - 1\}. \quad (21)$$

Then

$$Y_M \stackrel{d}{=} \frac{2Z_M - (M - 1)}{M}. \quad (22)$$

Let  $\eta_M = \text{Law}(Z_M)$ . Removing the first digit gives the exact self-similarity

$$Z_M \stackrel{d}{=} \frac{J + Z'_M}{2}, \quad J \perp Z'_M, \quad (23)$$

and therefore

$$\eta_M = \frac{1}{M} \sum_{j=0}^{M-1} (S_j)_* \eta_M, \quad S_j(x) = \frac{x+j}{2}. \quad (24)$$

The interval images

$$S_j([0, M-1]) = \left[ \frac{j}{2}, \frac{j+M-1}{2} \right]$$

form an overlapping chain whose union is  $[0, M-1]$ .

**Proposition 4.1** (Support, uniqueness, and atomlessness). *For every  $M \geq 2$ ,  $\eta_M$  is the unique invariant probability measure of (24), has full support  $[0, M-1]$ , and has no atoms. Consequently its CDF is continuous and strictly increasing.*

*Proof.* Uniqueness is the standard contraction argument in the Wasserstein metric. The support is the unique nonempty compact set fixed by the union of the maps  $S_j$ , and the interval calculation above identifies it as  $[0, M-1]$ .

For atomlessness, suppose the maximal atomic mass is  $a > 0$ , attained at  $x$ . From (24),

$$a = \eta_M\{x\} = \frac{1}{M} \sum_{j=0}^{M-1} \eta_M\{2x-j\}.$$

Every summand is at most  $a$ , so every summand must equal  $a$ . Iterating, all points

$$2^n x - \sum_{r=1}^n 2^{n-r} j_r, \quad 0 \leq j_r < M,$$

are atoms of mass  $a$ . The integer sums fill every integer from 0 to  $(M-1)(2^n-1)$ : this follows by induction because the translated intervals of attainable sums overlap. Hence the number of distinct atoms of mass  $a$  tends to infinity, a contradiction.  $\square$

## 4.2 The multiplicative cocycle

The quotient definition immediately yields an arithmetic semigroup law.

**Theorem 4.2** (Multiplicative cocycle). *For all integers  $M, N \geq 1$ ,*

$$Q_{MN}(z) = Q_M(z) Q_N(z/M). \quad (25)$$

*Equivalently, for independent copies,*

$$Y_{MN} \stackrel{d}{=} Y_M + \frac{1}{M} Y'_N, \quad (26)$$

*and hence*

$$Y_M + \frac{1}{M} Y'_N \stackrel{d}{=} Y_N + \frac{1}{N} Y'_M. \quad (27)$$

*Proof.* Factor the quotient:

$$\frac{\Phi(z)}{\Phi(z/(MN))} = \frac{\Phi(z)}{\Phi(z/M)} \frac{\Phi(z/M)}{\Phi(z/(MN))}.$$

The second factor is  $Q_N(z/M)$ . Characteristic-function uniqueness gives the random variable statements.  $\square$

**Corollary 4.3** (Radix- $M$  innovation expansion). *For each  $M \geq 2$ , if  $Y_M^{(0)}, Y_M^{(1)}, \dots$  are independent copies of  $Y_M$ , then*

$$X \stackrel{d}{=} \sum_{k=0}^{\infty} \frac{Y_M^{(k)}}{M^k}. \quad (28)$$

More generally,

$$Y_{M^k} \stackrel{d}{=} \sum_{j=0}^{k-1} \frac{Y_M^{(j)}}{M^j}, \quad Q_{M^k}(z) = \prod_{j=0}^{k-1} Q_M(z/M^j). \quad (29)$$

*Proof.* Iterate (14); the remainder after  $k$  steps is  $M^{-k} X^{(k)}$  and converges uniformly to zero. The finite statement is [theorem 4.2](#) iterated.  $\square$

Thus the same  $C^\infty$  law admits infinitely many exact innovation systems. For  $M = 2$  the innovations are uniform. For odd  $M$  they will be singular continuous. For even non-powers of two they are finitely smooth. This is a sharp arithmetic contrast inside one fixed limiting law.

### 4.3 Cocycle factorization by the 2-adic valuation

Write

$$M = 2^r q, \quad r = \nu_2(M), \quad q \text{ odd}. \quad (30)$$

Then [theorem 4.2](#) gives

$$Q_M(z) = Q_{2^r}(z) Q_q(z/2^r), \quad Y_M \stackrel{d}{=} Y_{2^r} + 2^{-r} Y'_q. \quad (31)$$

For the power-of-two factor, the original sinc product telescopes:

$$Q_{2^r}(z) = \prod_{j=0}^{r-1} \text{sinc}_{\text{rad}}\left(\frac{\pi z}{2^j}\right). \quad (32)$$

Hence  $Y_{2^r}$  is a sum of  $r$  independent centered uniforms with half-widths  $2^{-1}, 2^{-2}, \dots, 2^{-r}$ . Its density is a compact unequal-knot B-spline of degree  $r - 1$ .

## 5 Singular-to-smooth parity and exact regularity

### 5.1 A pure-type lemma for the digit measures

The following short argument is included to avoid using the Jessen–Wintner pure-type theorem [6] as a black box, although it is an instance of that classical principle.

**Lemma 5.1** (Pure Lebesgue type). *For every  $M \geq 2$ , the self-similar measure  $\eta_M$  is either purely absolutely continuous or purely singular with respect to Lebesgue measure.*

*Proof.* Write the Lebesgue decomposition  $\eta_M = f \, dx + \sigma$ . The affine operator

$$\mathcal{T}\rho = \frac{1}{M} \sum_{j=0}^{M-1} (S_j)_* \rho$$

preserves absolute continuity and singularity. Uniqueness of the Lebesgue decomposition in  $\eta_M = \mathcal{T}\eta_M$  therefore gives  $f \, dx = \mathcal{T}(f \, dx)$  and  $\sigma = \mathcal{T}\sigma$ . If both parts had positive mass, normalizing them would produce two distinct invariant probability measures for the contracting IFS, contradicting uniqueness.  $\square$

## 5.2 Odd divisors are singular continuous

Define the entire one-level ratio

$$R_M(z) = \frac{\text{sinc}_{\text{rad}}(\pi z)}{\text{sinc}_{\text{rad}}(\pi z/M)}. \quad (33)$$

Then

$$Q_M(z) = R_M(z)Q_M(z/2), \quad Q_M(2z) = R_M(2z)Q_M(z). \quad (34)$$

At a common zero  $z = M\ell$  the removable value is

$$R_M(M\ell) = (-1)^{(M-1)\ell}. \quad (35)$$

For odd  $M$  it is always 1.

**Theorem 5.2** (Odd- $M$  singularity and non-Rajchman subsequence). *If  $M \geq 3$  is odd, then  $R_M^{\text{law}}$  and  $\eta_M$  are singular continuous measures with full interval support. Moreover,*

$$Q_M(M2^k) = Q_M(M) \neq 0 \quad (k = 0, 1, 2, \dots). \quad (36)$$

*In particular,  $R_M^{\text{law}}$  is not Rajchman.*

*Proof.* By (34) and (35), doubling an argument of the form  $M2^k$  leaves  $Q_M$  unchanged. The value at  $M$  is nonzero: each factor in the digit product is nonzero, and the product converges normally. An absolutely continuous finite measure has a Fourier transform tending to zero at infinity, so  $R_M^{\text{law}}$  is not absolutely continuous. By [theorem 5.1](#) it is singular, and by [theorem 4.1](#) it is continuous with full interval support.  $\square$

This is an explicit full-support singular measure, not a Cantor-support measure. Its singularity comes from exact dyadic arithmetic resonances rather than from gaps in the support.

## 5.3 Even divisors and a complete regularity trichotomy

Let  $b_r$  denote the density of  $Y_{2^r}$  for  $r \geq 1$ . It is the convolution of the  $r$  box densities on  $[-2^{-j}, 2^{-j}]$ ,  $1 \leq j \leq r$ . Therefore

$$D^r b_r \quad \text{is a finite signed atomic measure,} \quad (37)$$

and  $b_r \in C^{r-2}$  but  $b_r \notin C^{r-1}$  for  $r \geq 2$ ;  $b_1$  is the indicator density of  $[-1/2, 1/2]$ .

**Theorem 5.3** (Exact arithmetic regularity classification). *Let  $M = 2^r q \geq 2$  with  $q$  odd.*

1. *If  $r = 0$  (equivalently,  $M$  is odd), then  $R_M^{\text{law}}$  is singular continuous.*
2. *If  $r \geq 1$  and  $q = 1$ , then  $R_M^{\text{law}}$  is the finite B-spline law  $R_{2^r}^{\text{law}}$ . For  $r = 1$  its density has jump discontinuities; for  $r \geq 2$  its density is exactly  $C^{r-2}$  and not  $C^{r-1}$ .*
3. *If  $r \geq 1$  and  $q > 1$ , then  $R_M^{\text{law}}$  has a compactly supported density  $g_M$  that is exactly*

$$g_M \in C^{r-1}(\mathbb{R}) \setminus C^r(\mathbb{R}). \quad (38)$$

*Its  $r$ th distributional derivative is a nonzero finite singular continuous signed measure.*

*Proof.* The first claim is [theorem 5.2](#); the second is standard finite convolution of boxes. For the third, (31) writes

$$R_M^{\text{law}} = R_{2^r}^{\text{law}} * \mathcal{D}_{2^{-r}} R_q^{\text{law}}.$$

Convolution with the B-spline density  $b_r$  gives an absolutely continuous law. Since  $D^r b_r$  is atomic,

$$D^r g_M = (D^r b_r) * \mathcal{D}_{2^{-r}} R_q^{\text{law}}$$

is a finite linear combination of translates of the atomless measure  $\mathcal{D}_{2^{-r}} \mathbf{R}_q^{\text{law}}$ . It has no atoms. A distribution whose derivative is an atomless finite measure is a continuous function of bounded variation; iterating backward shows  $g_M \in C^{r-1}$ .

It remains to prove that  $g_M \notin C^r$ . The leftmost atom of  $D^r b_r$  is unique and nonzero. The distance to the next spline knot is  $2^{1-r}$ , whereas the full width of the scaled odd core is

$$2 \cdot 2^{-r} \frac{q-1}{q} = 2^{1-r} \frac{q-1}{q} < 2^{1-r}.$$

Hence on a small interval adjacent to the left edge of the support,  $D^r g_M$  is a nonzero constant multiple of one translate of the singular measure  $\mathbf{R}_q^{\text{law}}$ , with no cancellation from other translates. Thus  $D^r g_M$  is not absolutely continuous there. If  $g_M$  were  $C^r$ , its  $r$ th distributional derivative would be a continuous function times Lebesgue measure, a contradiction.  $\square$

**Example 5.4.** The theorem gives the concrete ladder

| $M$             | type of $\mathbf{R}_M^{\text{law}}$                          |
|-----------------|--------------------------------------------------------------|
| 3, 5, 7, ...    | singular continuous, full interval support                   |
| 2               | box density                                                  |
| 4               | $C^0 \setminus C^1$ B-spline density                         |
| 6, 10, 14, ...  | $C^0 \setminus C^1$ density with singular derivative         |
| 8               | $C^1 \setminus C^2$ B-spline density                         |
| 12, 20, 28, ... | $C^1 \setminus C^2$ density with singular second derivative. |

The same 2-adic valuation gives one extra classical derivative in the non-power-of-two even case because the odd core is atomless rather than atomic.

## 5.4 A torus aliasing dichotomy

The integer Fourier samples will be computed exactly in [section 7](#); the result can already be stated in measure-theoretic form.

**Theorem 5.5** (Fractional-part law). *Let  $Y_M \bmod 1$  denote the pushforward of  $\mathbf{R}_M^{\text{law}}$  to  $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ .*

1. *If  $M$  is even,  $Y_M \bmod 1$  is Haar-uniform on  $\mathbb{T}$ .*
2. *If  $M$  is odd, its Fourier spectrum is supported on  $M\mathbb{Z}$ , it is invariant under translation by  $1/M$ , and it is not Haar-uniform.*

*If  $M$  is even and  $g_M$  is a density, then*

$$\sum_{k \in \mathbb{Z}} g_M(x+k) = 1 \tag{39}$$

*in  $L^1(\mathbb{T})$ , and pointwise whenever the periodization is continuous.*

*Proof.* By [theorem 7.1](#) below, for even  $M$  all nonzero integer Fourier coefficients vanish, so the torus law is Haar. For odd  $M$ , precisely the coefficients at multiples of  $M$  may survive, and the coefficient at  $M$  is nonzero by (36). Fourier support in  $M\mathbb{Z}$  is equivalent to invariance under translation by  $1/M$ . The density periodization has exactly these integer Fourier coefficients, proving (39) by Fourier uniqueness.  $\square$

This result places the divisor family directly in the repository's sampling and Strang–Fix landscape: even divisors reproduce constants under integer shifts, while odd divisors retain an exact  $M$ -fold alias.

## 6 Transport convergence, singular approximants, and the inverse Fabius function

### 6.1 Exact $W_\infty$ distance

The coupling in (14) gives more than weak convergence.

**Theorem 6.1** (Exact transport distance). *Let  $\mu = \text{Law}(X)$  and  $\mathbf{R}_M^{\text{law}} = \text{Law}(Y_M)$ . Then*

$$W_\infty(\mu, \mathbf{R}_M^{\text{law}}) = \frac{1}{M}. \quad (40)$$

For every  $1 \leq p < \infty$ ,

$$W_p(\mu, \mathbf{R}_M^{\text{law}}) \leq \frac{1}{M} (\mathbb{E}|X|^p)^{1/p}. \quad (41)$$

Consequently  $\mathbf{R}_M^{\text{law}} \Rightarrow \mu$  as  $M \rightarrow \infty$ . Along odd  $M$  this is convergence of singular continuous full-support measures to the  $C^\infty$  density up.

*Proof.* Under the natural coupling  $X = Y_M + X'/M$ , one has  $|X - Y_M| \leq 1/M$  almost surely, proving the upper bounds. The supports have respective left endpoints  $-1$  and  $-(M-1)/M$ . Quantiles approach these endpoints as  $p \downarrow 0$ , so no coupling can have essential displacement smaller than their difference  $1/M$ .  $\square$

**Corollary 6.2** (Maximal total variation along odd approximants). *For every odd  $M \geq 3$ ,  $\mu$  and  $\mathbf{R}_M^{\text{law}}$  are mutually singular. In the convention  $d_{\text{TV}}(\rho, \sigma) = \sup_A |\rho(A) - \sigma(A)|$ ,*

$$d_{\text{TV}}(\mu, \mathbf{R}_M^{\text{law}}) = 1, \quad W_\infty(\mu, \mathbf{R}_M^{\text{law}}) = \frac{1}{M}. \quad (42)$$

This gives a canonical arithmetic example in which transport convergence is exact and rapid while total variation convergence fails maximally.

### 6.2 Certified quantile approximants

Let  $G_X^{-1}$  and  $G_M^{-1}$  be the left-continuous quantile functions of  $X$  and  $Y_M$ . One-dimensional  $W_\infty$  equals the supremum norm of quantiles, so

$$\sup_{0 < p < 1} |G_X^{-1}(p) - G_M^{-1}(p)| = \frac{1}{M}. \quad (43)$$

By (4),  $G_X^{-1}(p) = 2Q_F(p) - 1$ . If  $K_M$  is the quantile function of  $Z_M$ , then (22) gives

$$G_M^{-1}(p) = \frac{2K_M(p) - (M-1)}{M}.$$

**Corollary 6.3** (Uniform inverse-Fabius bracket). *For every integer  $M \geq 2$  and every  $0 < p < 1$ ,*

$$\left| Q_F(p) - \left( \frac{K_M(p)}{M} + \frac{1}{2M} \right) \right| \leq \frac{1}{2M}. \quad (44)$$

The constant is best possible uniformly in  $p$ .

This approximation is complementary to the repository's endpoint Lambert- $W$  expansions. The Lambert analysis is asymptotically sharp as  $p \downarrow 0$ ; (44) is global in  $p$ , certified, and indexed by an arithmetic self-similar quantile.

### 6.3 Iterated and radix- $M$ quantile schemes

Let

$$S_{M,k} = \sum_{j=0}^{k-1} M^{-j} Y_M^{(j)} \stackrel{d}{=} Y_{M^k}. \quad (45)$$

Then  $X = S_{M,k} + M^{-k} X^{(k)}$  in the natural coupling. Hence

$$W_\infty(\text{Law}(S_{M,k}), \mu) = M^{-k}. \quad (46)$$

If  $L_{M,k}$  is the quantile of  $S_{M,k}$ ,

$$\left| Q_F(p) - \frac{1 + L_{M,k}(p)}{2} \right| \leq \frac{1}{2M^k}. \quad (47)$$

For  $M = 2^r$  each innovation is a finite spline sum, so this becomes a purely spline-based inverse approximation. For odd  $M$ , each finite approximation remains singular, yet its quantile converges uniformly to the inverse Fabius quantile.

### 6.4 A $q = M^{-2}$ Richardson hierarchy

The characteristic function of  $S_{M,k}$  is

$$P_{M,k}(z) = Q_{M^k}(z) = \frac{\Phi(z)}{\Phi(z/M^k)}. \quad (48)$$

Near the origin,

$$\log \Phi(w) = \sum_{r \geq 1} \frac{\kappa_{2r}(X) (2\pi i w)^{2r}}{(2r)!}. \quad (49)$$

Therefore, with  $q = M^{-2}$ ,

$$P_{M,k}(z) = \Phi(z) \exp \left( - \sum_{r \geq 1} \frac{\kappa_{2r}(X) (2\pi i z)^{2r}}{(2r)!} q^{rk} \right). \quad (50)$$

On every compact  $z$ -set this is a convergent expansion in powers  $q^k$ . Define recursively

$$R_k^{(0)}(z) = P_{M,k}(z), \quad R_k^{(j)}(z) = \frac{R_{k+1}^{(j-1)}(z) - q^j R_k^{(j-1)}(z)}{1 - q^j}. \quad (51)$$

**Proposition 6.4** (Integer-scale  $q$ -acceleration). *For fixed  $J$  and locally uniformly in  $z$ ,*

$$R_k^{(J)}(z) = \Phi(z) + \mathcal{O}(q^{(J+1)k}) \quad (k \rightarrow \infty). \quad (52)$$

*The coefficients obtained by expanding (51) are Gaussian  $q$ -binomial coefficients with  $q = M^{-2}$ .*

*Proof.* Equation (50) has the form  $P_{M,k} = \Phi + \sum_{r \geq 1} c_r q^{rk}$ . The  $j$ th recursion annihilates the term  $q^{jk}$  without reintroducing lower powers. Induction gives the result. The closed coefficient formula is the standard factorization of the annihilator  $\prod_{j=1}^J (E - q^j)/(1 - q^j)$ , where  $E$  shifts  $k$  to  $k+1$ .  $\square$

The repository's  $q = 1/4$  dyadic extrapolation is the case  $M = 2$ . The divisor family shows that the same Bell–Bernoulli/ $q$ -binomial mechanism exists at every reciprocal integer scale.

## 7 The arithmetic zero divisor and its spectral zeta function

### 7.1 Exact zero multiplicities

Let

$$\delta_M(n) = \text{ord}_{z=n} Q_M(z), \quad n \in \mathbb{N}_{>0}. \quad (53)$$

The quotient has removable cancellations whenever  $M \mid n$ .

**Proposition 7.1** (Zero multiplicity formula). *For every  $M \geq 2$  and  $n \geq 1$ ,*

$$\delta_M(n) = 1 + \nu_2(n) - \mathbf{1}_{M \mid n}(1 + \nu_2(n/M)). \quad (54)$$

Equivalently,

$$\delta_M(n) = \begin{cases} 1 + \nu_2(n), & M \nmid n, \\ \nu_2(M), & M \mid n. \end{cases} \quad (55)$$

Thus:

- if  $M$  is odd, the positive zero set is  $\mathbb{N}_0 \setminus M\mathbb{N}_0$ ;
- if  $M$  is even, every positive integer is a zero;
- on multiples of  $M$ , the residual order is the constant  $\nu_2(M)$ .

*Proof.* Subtract the zero order of  $\Phi(z/M)$  from that of  $\Phi(z)$ . The denominator vanishes at  $z = n$  exactly when  $M \mid n$ . If so,  $\nu_2(n) = \nu_2(M) + \nu_2(n/M)$ , giving the second form.  $\square$

The entire quotient has the canonical product

$$Q_M(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{\delta_M(n)}. \quad (56)$$

For odd  $M$ , the zero lattice has periodic holes at  $M\mathbb{N}_0$ ; for even  $M$  it remains the full integer lattice but with a flattened multiplicity on multiples of  $M$ .

### 7.2 A closed spectral zeta function

Define the positive zero-divisor zeta function

$$\mathcal{Z}_M(s) = \sum_{n=1}^{\infty} \frac{\delta_M(n)}{n^s}, \quad \Re s > 1. \quad (57)$$

Since

$$\sum_{n \geq 1} \frac{1 + \nu_2(n)}{n^s} = \frac{\zeta(s)}{1 - 2^{-s}}, \quad (58)$$

subtraction of the  $M$ -dilated divisor is immediate.

**Theorem 7.2** (Zero zeta formula). *For  $\Re s > 1$ ,*

$$\boxed{\mathcal{Z}_M(s) = (1 - M^{-s}) \frac{\zeta(s)}{1 - 2^{-s}}.} \quad (59)$$

*This identity gives a meromorphic continuation to  $\mathbb{C}$ . Its residue at  $s = 1$  is*

$$\text{Res}_{s=1} \mathcal{Z}_M(s) = 2 \left(1 - \frac{1}{M}\right). \quad (60)$$

At the nonzero dyadic lattice points

$$s_k = \frac{2\pi i k}{\log 2}, \quad k \in \mathbb{Z} \setminus \{0\}, \quad (61)$$

the residue, when nonzero, is

$$\operatorname{Res}_{s=s_k} \mathcal{Z}_M(s) = \frac{(1 - M^{-s_k})\zeta(s_k)}{\log 2}. \quad (62)$$

*Proof.* From (54),

$$\mathcal{Z}_M(s) = \sum_{n \geq 1} \frac{1 + \nu_2(n)}{n^s} - \sum_{m \geq 1} \frac{1 + \nu_2(m)}{(Mm)^s},$$

which is (59). The residue computations follow from the simple pole of  $\zeta$  at 1 and from  $(1 - 2^{-s})' = \log 2 \cdot 2^{-s}$ .  $\square$

### 7.3 Power-of-two rigidity and complex dimensions

**Theorem 7.3** (Equivalent power-of-two Euler cancellations). *For an integer  $M \geq 2$ , the following are equivalent:*

1.  $M$  is a power of two;
2. the numerator  $1 - M^{-s}$  cancels every nonzero zero  $s_k$  of  $1 - 2^{-s}$ ;
3. the factor  $(1 - M^{-s})/(1 - 2^{-s})$  is a finite Dirichlet polynomial;
4. there exists an integer  $r \geq 1$  such that  $Q_M(z) = \prod_{j=0}^{r-1} \operatorname{sinc}_{\text{rad}}(\pi z/2^j)$ .

If  $M = 2^r$ , then

$$\frac{1 - M^{-s}}{1 - 2^{-s}} = \sum_{j=0}^{r-1} 2^{-js}, \quad Q_M(z) = \prod_{j=0}^{r-1} \operatorname{sinc}_{\text{rad}}(\pi z/2^j). \quad (63)$$

If  $M$  is not a power of two, the Euler numerator cancels none of the nonzero  $s_k$ ;  $\mathcal{Z}_M$  has a genuine pole at each such point for which  $\zeta(s_k) \neq 0$ .

*Proof.* If  $M = 2^r$ , the finite geometric sum and (32) give all four statements. Conversely, the numerator cancels a denominator zero  $s_k$  exactly when  $k \log_2 M \in \mathbb{Z}$ . Cancellation at one nonzero  $k$  makes  $\log_2 M$  rational, and unique factorization then forces the integer  $M$  to be a power of two. A finite Dirichlet polynomial is entire, so statement 3 implies these cancellations. Finally, if statement 4 holds, characteristic-function uniqueness identifies  $R_M^{\text{law}}$  with  $R_{2^r}^{\text{law}}$ ; comparing the support half-widths  $(M - 1)/M$  and  $(2^r - 1)/2^r$  gives  $M = 2^r$ .  $\square$

The poles  $s_k$  may be viewed as complex dimensions of the arithmetic zero divisor. They do not describe a fractal support—the support is always an interval—but rather a log-periodic obstruction in the multiplicity geometry. In section 10 the same power-of-two rigidity reappears in an exact endpoint phase.

### 7.4 Values at the origin and a zeta-regularized determinant

Write  $a = \log_2 M$ . The singularity at  $s = 0$  is removable, and expansion gives

$$\mathcal{Z}_M(0) = -\frac{1}{2} \log_2 M. \quad (64)$$

Using  $\zeta'(0) = -\frac{1}{2} \log(2\pi)$ ,

$$\mathcal{Z}'_M(0) = -\frac{a}{2} \log(2\pi) - \frac{a}{4} \log\left(\frac{2}{M}\right). \quad (65)$$

Thus the zeta-regularized product of the positive zeros, counted with multiplicity, is

$$\exp(-\mathcal{Z}'_M(0)) = (2\pi)^{a/2} \left(\frac{2}{M}\right)^{a/4}. \quad (66)$$

At negative integers the continuation is controlled by Bernoulli numbers through  $\zeta(-n) = (-1)^n B_{n+1}/(n+1)$ , providing a second Bernoulli bridge independent of the cumulant formulas in [section 8](#).

## 8 Bernoulli cumulants, Bell moments, and Appell cocycles

### 8.1 The discrete digit cumulants

The moment generating function of the symmetric digit  $D_M$  is

$$\mathbb{E}e^{tD_M} = \frac{1}{M} \frac{\sinh t}{\sinh(t/M)}. \quad (67)$$

The classical expansion

$$\log \frac{\sinh t}{t} = \sum_{r=1}^{\infty} \frac{2^{2r-1} B_{2r}}{r(2r)!} t^{2r} \quad (68)$$

therefore gives an exact Bernoulli-number formula.

**Proposition 8.1** (Digit and divisor cumulants). *All odd cumulants of  $D_M$ ,  $Y_M$ , and  $X$  vanish. For  $r \geq 1$ ,*

$$\kappa_{2r}(D_M) = \frac{2^{2r} B_{2r}}{2r} (1 - M^{-2r}), \quad (69)$$

$$\kappa_{2r}(Y_M) = \frac{2^{2r} B_{2r}}{2r(2^{2r} - 1)} (1 - M^{-2r}), \quad (70)$$

$$\kappa_{2r}(X) = \frac{2^{2r} B_{2r}}{2r(2^{2r} - 1)}. \quad (71)$$

In particular,

$$\text{Var}(Y_M) = \frac{1 - M^{-2}}{9}, \quad \kappa_4(Y_M) = -\frac{2}{225}(1 - M^{-4}). \quad (72)$$

*Proof.* Subtract the expansion (68) at  $t/M$  from that at  $t$  to obtain (69). The independent series (12) multiplies the  $2r$ th digit cumulant by  $\sum_{n \geq 0} 2^{-2r(n+1)} = (2^{2r} - 1)^{-1}$ . Letting  $M \rightarrow \infty$ , or directly using (2), gives (71).  $\square$

The factor  $1 - M^{-2r}$  is the cumulant-level shadow of the quotient  $\Phi(z)/\Phi(z/M)$ . It is also the exact finite-difference symbol that later produces the zero zeta factor  $1 - M^{-s}$ .

### 8.2 Bernoulli-polynomial digit moments

Faulhaber's formula yields a companion expression in Bernoulli polynomials. For  $r \geq 0$ ,

$$\mathbb{E}D_M^{2r} = \frac{2^{2r}}{M^{2r+1}(2r+1)} \left[ B_{2r+1}\left(\frac{M+1}{2}\right) - B_{2r+1}\left(\frac{1-M}{2}\right) \right]. \quad (73)$$

The self-similarity  $Y_M \stackrel{d}{=} (D_M + Y'_M)/2$  gives a rational triangular recurrence for its moments  $m_n^{(M)} = \mathbb{E}Y_M^n$ :

$$(2^n - 1)m_n^{(M)} = \sum_{k=0}^{n-1} \binom{n}{k} \mathbb{E}D_M^{n-k} m_k^{(M)}, \quad m_0^{(M)} = 1. \quad (74)$$

All odd moments vanish. This supplies an exact symbolic route independent of Fourier products.

### 8.3 Complete Bell-polynomial moments

Let  $\mathcal{B}_n^{\text{exp}}$  denote the complete exponential Bell polynomial. The moment-cumulant formula specializes to

$$\mathbb{E}Y_M^{2n} = \mathcal{B}_{2n}^{\text{exp}}(0, \kappa_2(Y_M), 0, \kappa_4(Y_M), \dots, 0, \kappa_{2n}(Y_M)). \quad (75)$$

For example,

$$\mathbb{E}Y_M^2 = \frac{1 - M^{-2}}{9}, \quad (76)$$

$$\mathbb{E}Y_M^4 = \frac{(1 - M^{-2})^2}{27} - \frac{2(1 - M^{-4})}{225}, \quad (77)$$

$$\mathbb{E}Y_M^6 = \frac{5(1 - M^{-2})^3}{243} - \frac{2(1 - M^{-2})(1 - M^{-4})}{135} + \frac{16(1 - M^{-6})}{3969}. \quad (78)$$

The limit  $M \rightarrow \infty$  recovers the Rvachev moments. Conversely, the  $M$ -dependence separates every cumulant by a single Euler factor, which can be useful for exact moment reconstruction from two or more divisors.

### 8.4 An Appell semigroup law

Define the Appell polynomials associated with  $Y_M$  by

$$\frac{e^{xt}}{\mathbb{E}e^{tY_M}} = \sum_{n=0}^{\infty} \mathcal{A}_n^{(M)}(x) \frac{t^n}{n!}. \quad (79)$$

The cocycle of moment generating functions is

$$M_{MN}(t) = M_M(t)M_N(t/M). \quad (80)$$

**Proposition 8.2** (Appell cocycle). *For all  $x_1, x_2 \in \mathbb{C}$ ,*

$$\mathcal{A}_n^{(MN)}\left(x_1 + \frac{x_2}{M}\right) = \sum_{j=0}^n \binom{n}{j} M^{-j} \mathcal{A}_{n-j}^{(M)}(x_1) \mathcal{A}_j^{(N)}(x_2). \quad (81)$$

*Proof.* Factor the generating function using (80):

$$\frac{e^{x_1 t}}{M_M(t)} \frac{e^{x_2(t/M)}}{M_N(t/M)},$$

and compare coefficients. □

This identity packages the Bernoulli/Bell algebra into the multiplicative integer semigroup. It suggests a divisor-indexed family of generalized Bernoulli polynomials whose multiplication law is not the classical one in the degree variable, but in the arithmetic parameter  $M$ .

## 9 Thue–Morse quotients, bounded binary partitions, and Stern’s sequence

### 9.1 Finite digit polynomials

For  $M \geq 2$  and  $N \geq 1$  define

$$A_{M,N}(x) = \prod_{j=0}^{N-1} \left(1 + x^{2^j} + \dots + x^{(M-1)2^j}\right) = \sum_{n \geq 0} c_{M,N}(n) x^n. \quad (82)$$

The coefficient  $c_{M,N}(n)$  counts representations

$$n = \sum_{j=0}^{N-1} d_j 2^j, \quad 0 \leq d_j < M. \quad (83)$$

It is also the exact mass numerator of the finite digit law:

$$\mathbb{P}(2^N Z_{M,N} = n) = M^{-N} c_{M,N}(n), \quad Z_{M,N} = \sum_{j=1}^N \frac{J_j}{2^j}. \quad (84)$$

For fixed  $n$ ,  $c_{M,N}(n)$  stabilizes once  $2^N > n$ ; call the stable value  $c_M(n)$ .

Let

$$T_N(x) = \prod_{j=0}^{N-1} (1 - x^{2^j}) = \sum_{n=0}^{2^N-1} \varepsilon_n x^n, \quad \varepsilon_n = (-1)^{\text{wt}_2(n)}, \quad (85)$$

be the finite Thue–Morse polynomial.

**Theorem 9.1** (Exact Thue–Morse quotient identity). *For every  $M \geq 2$  and  $N \geq 1$ ,*

$$\boxed{A_{M,N}(x) T_N(x) = T_N(x^M)}. \quad (86)$$

*Coefficientwise in the formal power-series ring  $\mathbb{Z}[[x]]$ ,*

$$A_M(x) T(x) = T(x^M), \quad (87)$$

where

$$A_M(x) = \sum_{n \geq 0} c_M(n) x^n, \quad T(x) = \prod_{j \geq 0} (1 - x^{2^j}) = \sum_{n \geq 0} \varepsilon_n x^n. \quad (88)$$

*Equivalently,*

$$\sum_{k=0}^n c_M(k) \varepsilon_{n-k} = \begin{cases} \varepsilon_{n/M}, & M \mid n, \\ 0, & M \nmid n. \end{cases} \quad (89)$$

*Proof.* For each  $j$ ,

$$1 + x^{2^j} + \cdots + x^{(M-1)2^j} = \frac{1 - x^{M2^j}}{1 - x^{2^j}}.$$

Multiplying over  $j < N$  gives (86). Every coefficient of the infinite products depends on only finitely many factors, so the identity passes coefficientwise to  $\mathbb{Z}[[x]]$ . Comparing coefficients gives (89).  $\square$

This is a direct algebraic bridge between the positive digit counts underlying the Rvachev divisor and the signed Prouhet–Thue–Morse coefficients. It may be read as saying that  $A_M$  is the exact formal quotient of a dilated Thue–Morse product by the original one.

## 9.2 Carry recurrence

Separating the least significant digit gives

$$c_M(n) = \sum_{\substack{0 \leq j < M \\ j \equiv n \pmod{2}}} c_M\left(\frac{n-j}{2}\right), \quad c_M(0) = 1, \quad c_M(n) = 0 \ (n < 0). \quad (90)$$

This is a finite carry automaton. Its matrix products are natural candidates for the sharp fractional regularity and multifractal invariants posed later.

### 9.3 The $M = 3$ Stern/hyperbinary specialization

For  $M = 3$ , the digits 0, 1, 2 are precisely hyperbinary multiplicities: each power of two may be used at most twice. A classical theorem identifies the number of hyperbinary representations of  $n$  with Stern's diatomic sequence  $s(n+1)$  [9], where

$$s(0) = 0, \quad s(1) = 1, \quad s(2n) = s(n), \quad s(2n+1) = s(n) + s(n+1). \quad (91)$$

In the present notation this becomes

$$c_3(n) = s(n+1). \quad (92)$$

Indeed, (90) gives

$$c_3(2n) = c_3(n) + c_3(n-1), \quad c_3(2n+1) = c_3(n), \quad (93)$$

which is the shifted Stern recursion.

**Corollary 9.2** (Stern–Thue–Morse convolution). *In  $\mathbb{Z}[[x]]$ ,*

$$\left( \sum_{n \geq 0} s(n+1)x^n \right) \left( \sum_{n \geq 0} \varepsilon_n x^n \right) = \sum_{n \geq 0} \varepsilon_n x^{3n}. \quad (94)$$

*Equivalently,*

$$\sum_{k=0}^n s(k+1)\varepsilon_{n-k} = \begin{cases} \varepsilon_{n/3}, & 3 \mid n, \\ 0, & 3 \nmid n. \end{cases} \quad (95)$$

The same  $M = 3$  coefficients are boundary masses of the singular Rvachev divisor:

$$\mathbb{P}(2^N Z_{3,N} = n) = 3^{-N} c_{3,N}(n), \quad c_{3,N}(n) = s(n+1) \quad (2^N > n). \quad (96)$$

Thus Stern's sequence is not merely adjacent combinatorics; it gives exact finite-resolution mass numerators for a canonical singular convolution divisor of the up-function.

### 9.4 Power-of-two rationality versus nonpower Mahler structure

If  $M = 2^r$ , cancellation in (87) yields the rational series

$$A_{2^r}(x) = \frac{1}{\prod_{j=0}^{r-1} (1 - x^{2^j})}. \quad (97)$$

For non-powers of two, the quotient retains an infinite lacunary structure. This exactly parallels the finite-spline versus complex-dimension distinction of [theorem 7.3](#). A rigorous transcendence or natural-boundary classification of  $A_M$  is left as a frontier problem.

## 10 Exact endpoint log-periodicity and inverse scaling

### 10.1 CDF refinement and the endpoint phase

Let

$$H_M(x) = \eta_M((-\infty, x]) = \mathbb{P}(Z_M \leq x). \quad (98)$$

With the conventions  $H_M(x) = 0$  for  $x \leq 0$  and  $H_M(x) = 1$  for  $x \geq M-1$ , the IFS gives

$$H_M(x) = \frac{1}{M} \sum_{j=0}^{M-1} H_M(2x - j). \quad (99)$$

On the left endpoint interval  $0 \leq x \leq 1/2$ , only the digit  $j = 0$  contributes:

$$H_M(x) = \frac{1}{M} H_M(2x). \quad (100)$$

Put

$$\alpha_M = \log_2 M. \quad (101)$$

For  $x \in (0, 1/2]$ , write  $-\log_2 x = n + \theta$  with  $n \in \mathbb{N}_0$  and  $0 \leq \theta < 1$ . Iterating (100) gives

$$H_M(x) = x^{\alpha_M} \Omega_M(\theta), \quad \Omega_M(\theta) = 2^{\alpha_M \theta} H_M(2^{-\theta}). \quad (102)$$

Extending  $\Omega_M$  periodically makes this an exact, not asymptotic, log-periodic law.

**Theorem 10.1** (Endpoint phase rigidity). *The periodic phase  $\Omega_M$  in (102) is constant if and only if  $M$  is a power of two.*

*If  $M = 2^r$ , then for  $0 \leq x \leq 1$ ,*

$$H_{2^r}(x) = \frac{x^r}{r! 2^{r(r-1)/2}}, \quad (103)$$

*and hence*

$$\Omega_{2^r} \equiv \frac{1}{r! 2^{r(r-1)/2}}. \quad (104)$$

*If  $M$  is not a power of two,  $\Omega_M$  is nonconstant.*

*Proof.* Suppose first that  $M = 2^r$ . Transforming the uniform-sum representation of  $Y_{2^r}$  to  $Z_{2^r}$  shows

$$Z_{2^r} \stackrel{d}{=} V_0 + 2V_1 + \cdots + 2^{r-1}V_{r-1}, \quad V_j \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1]. \quad (105)$$

For  $0 \leq x \leq 1$  none of the upper box constraints is active. The volume of the weighted simplex is  $x^r/r!$  divided by the Jacobian  $1 \cdot 2 \cdots 2^{r-1} = 2^{r(r-1)/2}$ , proving (103).

Now write a non-power  $M$  as  $2^r q$  with odd  $q > 1$ . If  $r = 0$ , a constant phase would make  $H_M(x) = Cx^{\alpha_M}$  on  $(0, 1/2)$ , giving a nonzero absolutely continuous restriction of the purely singular measure  $\eta_M$ , contradicting theorem 5.2.

If  $r \geq 1$ , the density of  $\eta_M$  is an affine translate and scale of  $g_M$  from theorem 5.3. On a sufficiently short left-edge interval, its  $r$ th distributional derivative is a nonzero translate of the singular odd core, because the leftmost spline knot is isolated by more than the width of that core. A constant phase would again give  $H_M(x) = Cx^{\alpha_M}$ , whose density has an  $r$ th derivative proportional to  $x^{\alpha_M - r - 1} = x^{\log_2 q - 1}$ , a continuous locally integrable function. This contradicts the singular derivative measure.  $\square$

**Corollary 10.2** (Fourier-zeta-endpoint trichotomy). *For  $M \geq 2$ , the following are equivalent:*

1.  *$M$  is a power of two;*
2. *there exists  $r \geq 1$  such that  $Q_M(z) = \prod_{j=0}^{r-1} \text{sinc}_{\text{rad}}(\pi z/2^j)$ ;*
3. *the Euler numerator  $1 - M^{-s}$  cancels all nonzero dyadic denominator zeros;*
4. *the endpoint CDF phase  $\Omega_M$  is constant;*
5. *there exists  $r \geq 1$  such that the bounded-binary-partition series is  $A_M(x) = 1/\prod_{j=0}^{r-1}(1 - x^{2^j})$ .*

*Proof.* The first four equivalences combine theorems 7.3 and 10.1. The power-of-two case of the fifth is (97). Conversely, multiplying the asserted rational form by  $T(x)$  and using (87) gives  $T(x^M) = T(x^{2^r})$ . The first nonconstant terms of these formal series are respectively  $-x^M$  and  $-x^{2^r}$ , so  $M = 2^r$ .  $\square$

This is a rigidity statement spanning probability, entire functions, Dirichlet series, endpoint asymptotics, and digital combinatorics.

## 10.2 Quantile phase

By [theorem 4.1](#),  $H_M$  has a continuous inverse

$$K_M(p) = H_M^{-1}(p), \quad 0 < p < 1. \quad (106)$$

Equation (100) is equivalent to

$$K_M(p/M) = \frac{1}{2} K_M(p) \quad (107)$$

whenever  $K_M(p) \leq 1$ . Put

$$\beta_M = \log_M 2 = \frac{1}{\alpha_M}. \quad (108)$$

Then for sufficiently small  $p$ ,

$$K_M(p) = p^{\beta_M} \Psi_M(\log_M p), \quad (109)$$

where  $\Psi_M$  is positive, continuous, and 1-periodic.

**Corollary 10.3** (Inverse phase rigidity). *The phase  $\Psi_M$  is constant if and only if  $M$  is a power of two. In the power-of-two case  $M = 2^r$ ,*

$$K_M(p) = \left( r! 2^{r(r-1)/2} p \right)^{1/r} \quad (110)$$

through the endpoint range corresponding to  $K_M(p) \leq 1$ .

*Proof.* The scaling relation makes  $K_M(p)/p^{\beta_M}$  periodic in  $\log_M p$ . It is constant exactly when  $H_M$  is a pure power, so [theorem 10.1](#) applies.  $\square$

The phase here is elementary and exact, unlike the much subtler logarithmic modulation in the inverse Fabius endpoint expansion. Its role is nevertheless useful: it supplies a fully solvable model of how discrete scale invariance enters an inverse distribution function and it appears inside the certified global bracket (44).

## 11 Polynomial and Legendre transfer formulas

### 11.1 A finite polynomial smoothing operator

The decomposition  $X = Y_M + X'/M$  acts exactly on polynomials. For a polynomial  $p$ , define

$$(\mathcal{S}_M p)(y) = \mathbb{E} p\left(y + \frac{X}{M}\right) = \sum_{k=0}^{\deg p} \frac{m_k(X)}{k! M^k} p^{(k)}(y), \quad (111)$$

where  $m_k(X) = \mathbb{E} X^k$ . Then

$$\int p(x) \mu(dx) = \int (\mathcal{S}_M p)(y) \mathbf{R}_M^{\text{law}}(dy). \quad (112)$$

Because all odd moments of  $X$  vanish, only even derivatives occur.

**Proposition 11.1** (Legendre coefficient transfer). *For the Legendre polynomial  $P_n$ ,*

$$\mathbb{E} P_n(X) = \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{m_{2j}(X)}{(2j)! M^{2j}} \int_{\mathbb{R}} P_n^{(2j)}(y) \mathbf{R}_M^{\text{law}}(dy). \quad (113)$$

More generally, for any affine scaling  $P_n(ax + b)$  the same formula holds after the corresponding chain-rule factors.

*Proof.* Apply (111) to  $p = P_n$  and integrate against  $\mathbf{R}_M^{\text{law}}$ .  $\square$

Since  $P_n^{(k)}$  is a finite linear combination of Legendre polynomials of lower degree, (113) is a triangular transfer system between the Legendre moments of the smooth Rvachev law and those of its arithmetic divisor. When  $M = 2^r$ , the divisor is a finite spline and every right-hand integral reduces to elementary truncated powers. When  $M$  is odd, the same identity transfers smooth Legendre data to a singular measure.

## 11.2 Cocycle composition of polynomial smoothing

The random decomposition shows

$$\mathcal{S}_{MN} = \mathcal{S}_N^{(1/M)} \circ \mathcal{S}_M, \quad (114)$$

where  $\mathcal{S}_N^{(1/M)}$  means that the  $N$ -scale smoothing is applied after the argument and derivative scaling inherited from  $y + M^{-1}Y_N$ . In exponential generating form this is exactly the Appell identity (81). A useful future project is to express this cocycle directly in the Legendre basis and study the resulting lower triangular matrices as  $M$  ranges over primes.

## 11.3 A representation-theoretic interpretation

Let  $\mathcal{C}_\rho$  be convolution by a compactly supported measure  $\rho$ . Then (18) says

$$\mathcal{C}_{\mathbf{R}_M^{\text{law}}} \mathcal{D}_{1/M} \mu = \mu. \quad (115)$$

On exponentials,  $\mathcal{C}_{\mathbf{R}_M^{\text{law}}}$  has multiplier  $Q_M$ ; on polynomials, it is the finite Bell-moment operator above; on the integer torus, its aliasing behavior is governed by [theorem 5.5](#). The divisor family therefore supplies one operator with three exact matrix models: diagonal in Fourier space, triangular in polynomial/Legendre space, and arithmetic on lattice samples.

# 12 Numerical experiments and reproducibility

## 12.1 What was computed

The accompanying script `experiments.py` performs four types of checks:

1. exact integer verification of  $A_{M,N}(x)T_N(x) = T_N(x^M)$  for several  $M, N$ ;
2. exact verification of  $c_3(n) = s(n+1)$  through  $n = 2048$ ;
3. stable high-precision evaluation of  $Q_M(M2^k)$  from digit factors, avoiding removable zero-over-zero quotients;
4. finite-level digit cascades, endpoint phase profiles, and zero multiplicity plots.

All exact polynomial calculations use Python integers. Floating-point plots are diagnostic only and are not used as proofs.

## 12.2 Repository provenance and replay boundary

The repository package was filed from the rootless archive `fabius_rvachev_frontier_report_2026-08-30-B.zip`, whose outer SHA-256 is `cfae82f303c3740bd76673fed772b1f69b9fedb0a911505360c930db7cc5a13f`. This package-local normalization is anchored to the two repository merge parents `563b90f25e0563cc521a53e9b478b667eaf3b694` and `685e1963897b1ff2cff8f541e46218cf8d37405a`. These hashes define a provenance boundary;

they are not a claim that the numerical script was executed at either revision. No experiment was executed during this source-only normalization.

The group record in `./README.md` documents an earlier temp-isolated Python 3.12 replay. Four CSV files were byte-identical, the text summary was equivalent up to line endings, and `endpoint_profiles.csv` differed in 66 last-place fields, with maximum absolute difference  $1.1102230246251565 \times 10^{-16}$ . All four PNG files showed layout drift between the unpinned packaged Matplotlib 3.10.8 and replayed Matplotlib 3.11.1 (1475 versus 1476 pixels wide). The experiment therefore supplies scientifically stable diagnostics, not a promise of byte-identical plots across Matplotlib versions. The current writer fixes CSV line endings explicitly to LF, but the package still has no dependency lock.

### 12.3 Odd and even cascade behavior

At level  $N$ , spread each atom of  $Z_{M,N}$  uniformly across one dyadic box of width  $2^{-N}$ . The box height at  $n/2^N$  is

$$\rho_{M,N}(n/2^N) = \left(\frac{2}{M}\right)^N c_{M,N}(n). \quad (116)$$

For odd  $M = 3$ , increasingly fine approximants develop persistent dyadic spikes, consistent with the proved singular continuous limit.

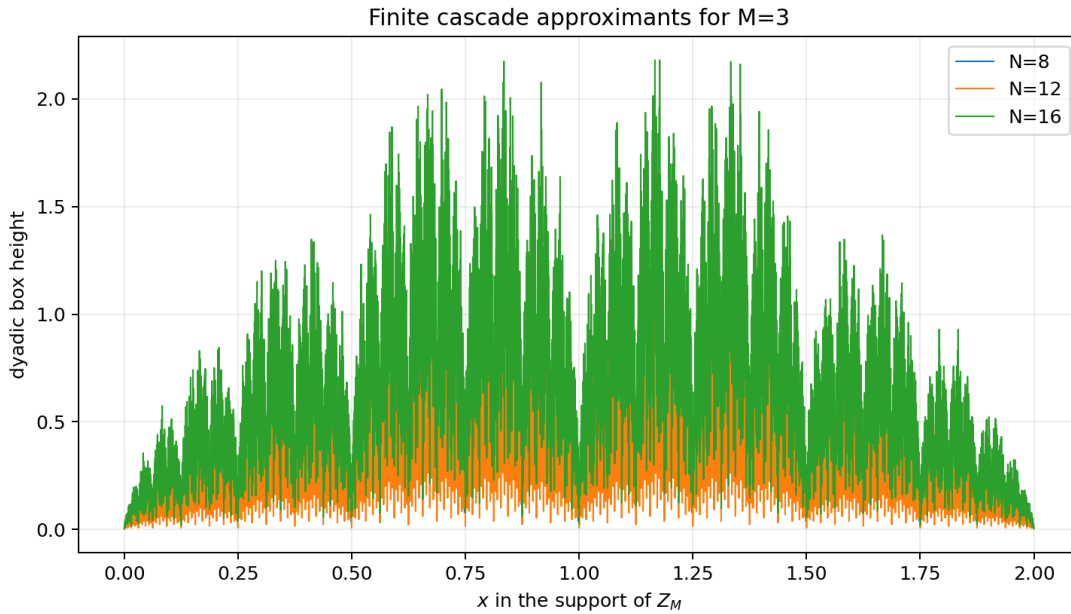


Figure 1: Dyadic box approximants for  $Z_3$  at levels  $N = 8, 12, 16$ . The plot is a visualization of finite atomic masses, not a claim that the singular limit has a density.

For  $M = 6 = 2 \cdot 3$ , the uniform factor  $R_2^{\text{law}}$  smooths the singular ternary core. The finite cascades rapidly agree with the continuous  $C^0 \setminus C^1$  density predicted by [theorem 5.3](#).

The exact output file `data/cascade_summary.csv` records maximum box heights and  $L^2$  energies. For  $M = 3$ , the maximum grows from approximately 1.3266 at  $N = 8$  to 2.4313 at  $N = 16$ , while for  $M = 6$  the profiles stabilize near the bounded continuous limit.

### 12.4 Endpoint phases

The numerical phase

$$\Omega_{M,N}(\theta) = 2^{(\log_2 M)\theta} \mathbb{P}(Z_{M,N} \leq 2^{-\theta}) \quad (117)$$

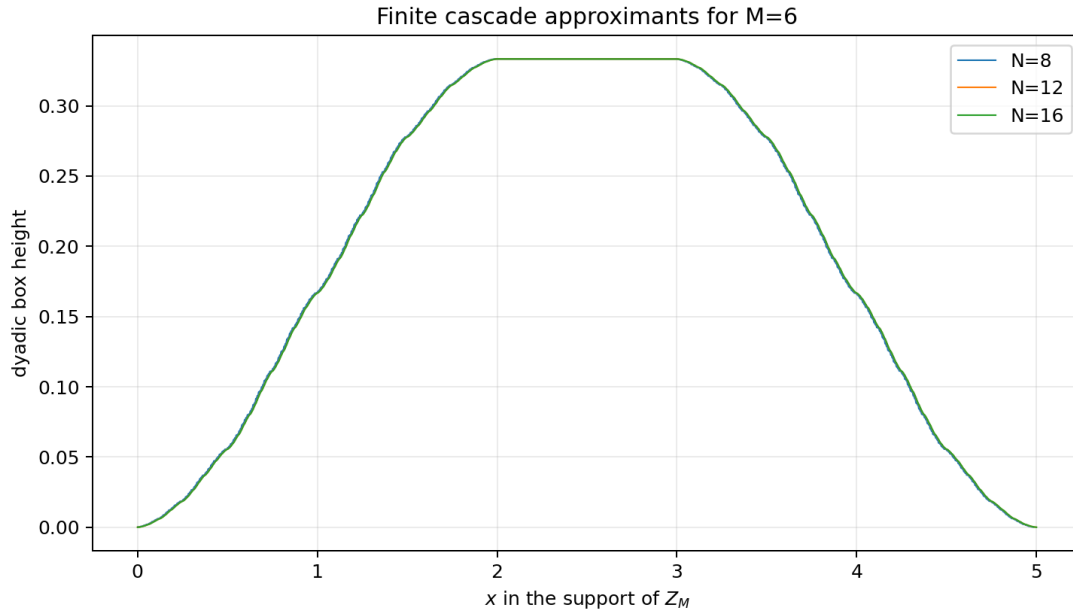


Figure 2: Dyadic box approximants for  $Z_6$ . The near-coincidence of the three levels is consistent with absolute continuity; the theorem, not the picture, identifies exact regularity  $C^0 \setminus C^1$ .

approximates the exact  $\Omega_M$ . The  $M = 4$  curve is the constant  $1/4$  from (104); non-power cases show nonconstant phases, sometimes with small amplitude.

### 12.5 Zero multiplicities and Fourier resonance

The high-precision calculation gives

$$|Q_3(3 \cdot 2^k)| = 0.083432097593273364307 \dots,$$

$$|Q_5(5 \cdot 2^k)| = 0.015632569447946944078 \dots,$$

for every tested  $0 \leq k \leq 10$ , exactly as predicted by (36). The values and thirty-digit real parts are in data/odd\_fourier\_subsequence.csv.

### 12.6 Reproduction command

From the extracted archive directory, run

```
python experiments.py --output-dir .
```

The script regenerates all four PNG figures and all CSV/text data files. Its final exact checks raise an exception if either the Thue–Morse quotient identity or the Stern equality fails.

## 13 Mellin complex dimensions of the endpoint phase

The exact edge law also has a short Mellin formulation. Put

$$\mathfrak{M}_M(s) = \int_0^{1/2} H_M(x) x^{s-1} dx, \quad B_M(s) = \int_{1/2}^1 H_M(x) x^{s-1} dx. \quad (118)$$

The first integral converges initially for  $\Re s > -\alpha_M$ , while  $B_M$  is entire.

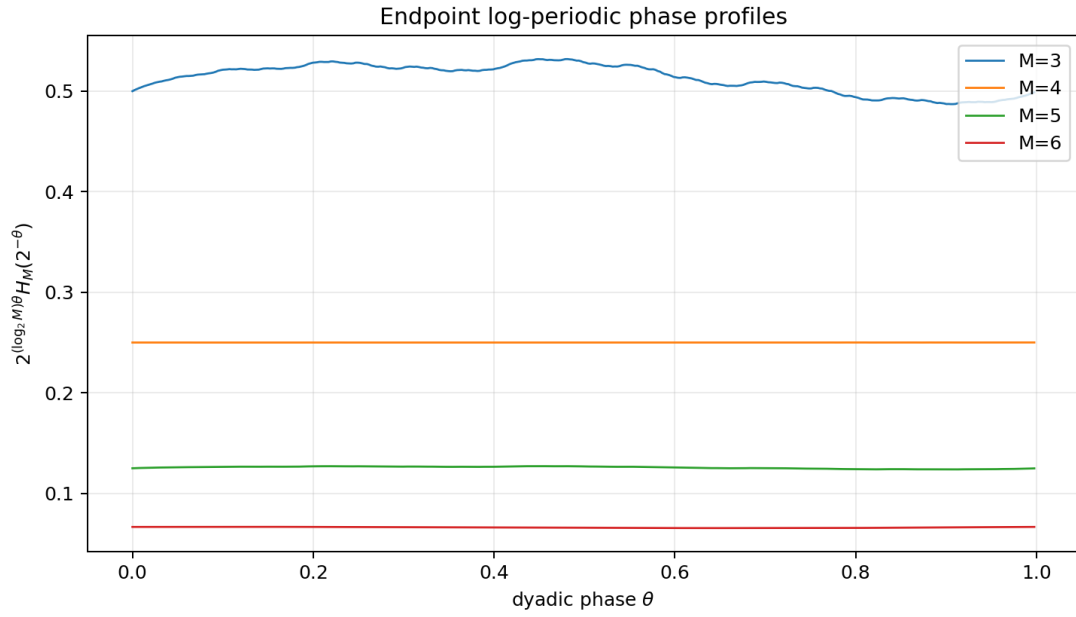


Figure 3: Finite-level endpoint phase profiles. Power-of-two  $M = 4$  is exactly constant;  $M = 3, 5, 6$  are nonconstant by [theorem 10.1](#).

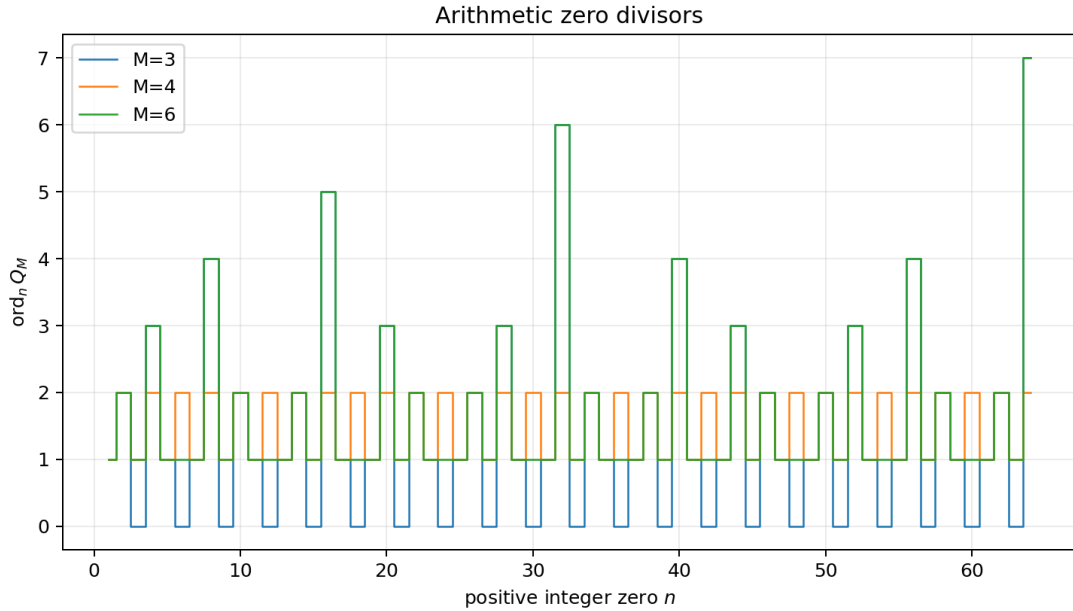


Figure 4: The first arithmetic zero multiplicities for  $M = 3, 4, 6$ . The  $M = 3$  divisor has holes at multiples of 3; even  $M$  retain every integer zero.

**Proposition 13.1** (Endpoint Mellin lattice). *The endpoint Mellin transform has the meromorphic continuation*

$$\mathfrak{M}_M(s) = \frac{B_M(s)}{M2^s - 1}. \quad (119)$$

Its candidate pole lattice is

$$s = -\log_2 M + \frac{2\pi i k}{\log 2}, \quad k \in \mathbb{Z}. \quad (120)$$

The residue at a noncancelled pole is  $B_M(s)/\log 2$ . All nonzero-frequency poles cancel if and only if  $M$  is a power of two; for every non-power of two at least one nonreal pole survives.

*Proof.* Use (100) and the substitution  $u = 2x$ :

$$\mathfrak{M}_M(s) = \frac{2^{-s}}{M} \int_0^1 H_M(u) u^{s-1} du = \frac{2^{-s}}{M} (\mathfrak{M}_M(s) + B_M(s)).$$

Solving gives (119). The pole lattice and residues are immediate. To identify cancellations, decompose  $(0, 1/2]$  into dyadic shells and put  $x = 2^{-(n+\theta)}$ . Up to a nonzero constant independent of  $k$ , the residue at  $s = -\alpha_M + 2\pi i k / \log 2$  is the Fourier coefficient  $\int_0^1 \Omega_M(\theta) e^{-2\pi i k \theta} d\theta$ . Hence a constant phase cancels every nonzero-frequency pole, whereas a nonconstant continuous periodic phase leaves at least one such pole. Apply [theorem 10.1](#).  $\square$

The zero-zeta lattice in (61) and the endpoint lattice in (120) differ by the real shift  $-\log_2 M$ . This is exactly what one expects when a log-periodic function is multiplied by the leading power  $x^{\log_2 M}$ . The result gives a precise bridge between the arithmetic zero divisor and the inverse-function phase, while stopping short of claiming that their residues coincide.

## 14 Conjectures and frontier problems

The proved theorems leave several sharply formulated questions. The conjectures below are chosen to be falsifiable and computationally accessible.

### 14.1 Fractional regularity from carry matrices

For  $M = 2^r q$  with odd  $q > 1$ , [theorem 5.3](#) determines the exact integer regularity  $C^{r-1} \setminus C^r$  but not the modulus of continuity of the highest continuous derivative. The carry recurrence (90) yields a finite set of transition matrices acting on overlap states.

**Conjecture 14.1** (Carry spectral radius controls sharp Hölder regularity). *For each odd  $q > 1$  there is an explicitly constructible finite family of nonnegative carry matrices  $\mathcal{C}_q$  such that the global Hölder exponent of the CDF of  $R_q^{\text{law}}$ , and therefore the Hölder exponent of  $g_{2^r q}^{(r-1)}$ , is a simple logarithmic function of the joint spectral radius of  $\mathcal{C}_q$ . The same matrices determine the optimal Besov and Sobolev exponents of  $g_{2^r q}$ .*

A practical route is to build automata for pairs of digit strings with equal or neighboring binary values, then study collision energies

$$E_{q,N} = \sum_n c_{q,N}(n)^2.$$

The growth exponent of  $E_{q,N}$  should give the  $L^2$  dimension and a first spectral bound for regularity. The included code already outputs finite-level energies.

## 14.2 Infinitely rich endpoint phases

**Conjecture 14.2** (No finite Fourier phase outside the spline cases). *If  $M$  is not a power of two, the periodic functions  $\Omega_M$  and  $\Psi_M$  have infinitely many nonzero Fourier coefficients. Equivalently, the endpoint Mellin transform (119) has infinitely many noncancelled nonreal poles.*

A stronger version would estimate the decay of the phase coefficients from the exact regularity class in [theorem 5.3](#). This would turn the qualitative complex-dimension picture into an explicit oscillatory asymptotic calculus.

## 14.3 Dyadic arithmetic of the divisor CDF

The original Fabius function takes rational values at dyadic arguments. The digit measures have even more explicit rational transition probabilities but also exact overlaps.

**Conjecture 14.3** (Rational dyadic values). *For every  $M \geq 2$ , every integer  $n \geq 0$ , and every integer  $k$ ,*

$$H_M(k/2^n) \in \mathbb{Q}. \quad (121)$$

*Moreover, these values can be computed by a finite carry Markov chain with a rational transition matrix whose dimension is  $O(M)$ , and their reduced denominators satisfy an explicit  $M$ -adic/2-adic valuation formula.*

For  $M = 3$ , the boundary layers already involve Stern numbers. A successful proof should connect the finite Markov resolvent to Stern–Brocot or Calkin–Wilf matrices.

## 14.4 Stern multifractality inside the singular $M = 3$ divisor

**Conjecture 14.4** (Stern spectrum). *The local dimension spectrum of  $\mathbb{R}_3^{\text{law}}$  is the Legendre transform of a pressure function built from the standard two Stern matrices. In particular, extremal local dimensions occur at eventually periodic binary addresses and are logarithms of quadratic units divided by  $\log 2$ .*

The exact mass formula (96) makes this accessible: cylinder and near-cylinder masses are controlled by products of the two matrices generating the Stern recursion. This appears to be a direct bridge from the Rvachev function to the metric theory of the Calkin–Wilf tree.

## 14.5 Mahler and natural-boundary rigidity

**Conjecture 14.5** (Natural boundary for nonpower digit quotients). *If  $M$  is not a power of two, the power series*

$$A_M(x) = \frac{T(x^M)}{T(x)} \quad (122)$$

*has the unit circle as a natural boundary and is transcendental over  $\mathbb{C}(x)$ . For algebraic  $0 < |\xi| < 1$ , broad classes of values  $A_M(\xi)$  are transcendental by a suitable Mahler method.*

The power-of-two cases are exactly rational by (97); hence this conjecture would add a function-theoretic member to the rigidity equivalences in [theorem 10.2](#).

## 14.6 General integer bases

For an integer base  $b \geq 2$ , consider

$$\Phi_b(z) = \prod_{n=0}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{\pi z}{b^n}\right). \quad (123)$$

The same Dirichlet-kernel calculation constructs  $Q_{b,M}(z) = \Phi_b(z)/\Phi_b(z/M)$  for every integer  $M$ .

**Conjecture 14.6** (General-base divisor classification). *For every integer  $b \geq 2$ :*

1. *the only independent-copy scales of the  $\Phi_b$  law are 0 and  $\pm 1/M$ ;*
2. *if  $M = b^r q$  with  $b \nmid q$ , then the divisor is singular continuous when  $r = 0$  and has exactly the regularity obtained by convolving the singular core with  $r$  box laws when  $r > 0$ ;*
3. *its zero-divisor zeta function is*

$$\mathcal{Z}_{b,M}(s) = (1 - M^{-s}) \frac{\zeta(s)}{1 - b^{-s}}; \quad (124)$$

4. *constant endpoint phase, finite sinc product, and  $M = b^r$  are equivalent.*

Most proof steps from the dyadic case transfer verbatim. The remaining care concerns composite-base valuations, knot separation, and the exact normalization of the generalized atomic function.

*Editorial note (ProveIt, 2026-09-29).* Item 1 is proved for prime and prime-power bases  $b$  in the later unreviewed draft `spectra-and-arithmetic/Simultaneous_Convolution_Divisors_Fabius_Type_Laws/` (2026-09-29), which recovers [theorem 3.3](#) as the single-copy case of a simultaneous-divisor criterion; for  $b$  prime and  $M = b^r$  its critical-regularity theorem gives the box-law (B-spline) case of item 2. Items 3 and 4, and item 2 when  $b \nmid M$ , remain open there. That draft writes the  $\Phi_b$  law as the law of  $X_b/2$  with  $X_b = \sum_{k \geq 0} b^{-k} U_k$  and  $U_k$  uniform on  $[-1, 1]$ .

## 14.7 Legendre spectra and arithmetic preconditioners

**Conjecture 14.7** (Divisor preconditioning of Legendre expansions). *For suitable  $M = M(n)$ , the triangular transfer (113) yields a better-conditioned computation of the high-degree Legendre coefficient  $\mathbb{E}P_n(X)$  than direct integration against  $\mu$ . Powers of two optimize symbolic simplicity, while even non-powers optimize a tradeoff between compact spline smoothing and nontrivial endpoint phase.*

A concrete experiment is to compare condition numbers of the lower triangular moment matrices for  $M = 2^r$ ,  $3 \cdot 2^r$ , and prime  $M$ , using exact Bernoulli cumulants. The cocycle suggests prime-factor preconditioners assembled multiplicatively.

## 14.8 Hybrid Lambert- $W$ /divisor inversion

The repository's inverse-Fabius expansion locates the extreme endpoint scale through the  $W_{-1}$  branch. The divisor bracket (44) is uniform but coarse unless  $M$  is large.

**Conjecture 14.8** (Hybrid certified inversion). *There is an adaptive scheme choosing  $M$  and an innovation depth  $k$  from the Lambert phase of  $p$  such that:*

1. *the Lambert- $W$  approximation provides a narrow initial interval for  $Q_F(p)$ ;*
2. *the quantile of  $Y_{M^k}$  refines it with the rigorous radius  $1/(2M^k)$ ;*
3.  *$q = M^{-2}$  Richardson extrapolation reduces the Fourier/cascade work to polylogarithmic complexity in the requested precision.*

This would combine two presently separate parts of the corpus: all-orders endpoint asymptotics and exact arithmetic self-similarity.

## 15 A formalization and research program

### 15.1 Current exact Lean adjacency and the quotient-family boundary

At the merge-parent boundary above, two exact Lean modules are adjacent to this report's zero-divisor program. In `Lean/FabiusFunction/GeneralizedZeroDivisor.lean`, `Fabius.generalizedRvachevProduct_eq_zero_iff_int` gives the arithmetic integer zero set for a generalized Rvachev product, while `Fabius.analyticOrderAt_generalizedRvachevProduct_int` gives its exact order at every nonzero integer. In `Lean/FabiusFunction/ReciprocalIntegerGammaZeros.lean`, `Fabius.geometricReciprocalGamma_inv_natCast_eq_zero_iff` identifies the zero set of the geometric reciprocal Gamma function at every reciprocal-integer base.

These declarations are exact formal infrastructure, but they concern generalized Rvachev products and geometric reciprocal Gamma functions. They do not formalize the extension of  $Q_M = \Phi/\Phi(\cdot/M)$  through its removable singularities, its characteristic-function interpretation, the reciprocal-integer convolution-divisor classification, the quotient multiplicity formula, or the spectral zeta function in this report. Those claims remain manuscript-level results; the following list is a proposed Lean development, not a crosswalk asserting that the quotient family is already formalized.

### 15.2 Lean-sized theorem units

The main arguments decompose naturally into formalization units:

1. prove the finite geometric-exponential identity (8);
2. construct  $Y_M$  as an almost surely convergent bounded random series and identify its characteristic function;
3. prove  $\Phi = \Phi(\cdot/M)Q_M$  and the decomposition theorem;
4. use the existing formal zero-order theorem for  $\Phi$  to prove the scale classification and (54);
5. formalize the cocycle and its random-series coupling;
6. formalize the finite polynomial identity (86);
7. derive cumulants as formal power-series identities before invoking analytic mgfs;
8. formalize the endpoint CDF scaling and power-of-two simplex volume;
9. isolate the measure-theoretic pure-type and regularity arguments.

The first seven are comparatively low risk because they reduce to finite sums, entire products already present in the repository, and integer valuations. The exact regularity statement needs the most measure/distribution infrastructure.

### 15.3 Computational priorities

The following experiments would most efficiently test the conjectures:

1. build minimal carry automata for  $M = 3, 5, 7, 9$  and compute joint spectral radii;
2. estimate Fourier coefficients of  $\Omega_M$  at high precision using exact recursive CDF evaluation rather than finite histograms;
3. solve rational linear systems for  $H_M(k/2^n)$  and search denominator patterns;
4. compute Legendre transfer matrices exactly through degree 100 and compare conditioning across factorizations of  $M$ ;
5. test natural-boundary behavior of  $A_M$  with Padé tables and Mahler functional relations;

6. compare odd singular and even smooth approximants in Wasserstein, Kolmogorov, negative Sobolev, and total variation metrics.

### 15.4 Conceptual payoff

The divisor family changes the usual viewpoint on up. Rather than one special dyadic infinite convolution, the Rvachev law becomes the terminal object of an arithmetic system of innovation laws. Prime factorization acts by convolution cocycles; 2-adic valuation acts as a regularity counter; odd factors generate singular cores; Thue–Morse signs invert the positive digit counts; Bernoulli numbers diagonalize cumulants; Bell polynomials reconstruct moments; and the inverse Fabius function is uniformly approximated by divisor quantiles. This unifies several themes of the repository without repeating its existing endpoint, Legendre, or  $q$ -series derivations.

## 16 Conclusion

The central object of this report is the entire quotient

$$Q_M(z) = \frac{\Phi(z)}{\Phi(z/M)}.$$

A one-line Dirichlet-kernel identity turns it into a characteristic function, but the consequences are unexpectedly broad. The zeros of  $\Phi$  classify all possible scales. Prime multiplication becomes a convolution cocycle. The odd part of  $M$  determines a singular continuous core, while the 2-adic valuation counts smoothing boxes. The same arithmetic appears in zero multiplicities, a closed zeta function, torus aliasing, endpoint log-periodicity, and finite digital polynomials. At  $M = 3$ , Stern’s sequence enters as the exact hyperbinary mass profile and is inverse-convolved by Thue–Morse signs. As  $M \rightarrow \infty$ , both smooth and singular divisors approach the Rvachev law at the exact transport rate  $1/M$ .

The strongest proved frontier statements are therefore not isolated formulas but a coherent arithmetic structure:

$$\boxed{\text{reciprocal scale} \iff \text{integer divisor} \iff \text{digit IFS} \iff \text{zero-zeta Euler factor.}}$$

The next decisive steps are to compute the carry spectra, resolve dyadic arithmetic of the divisor CDFs, formalize the classification, and connect Mellin phase residues to the repository’s all-orders inverse and asymptotic machinery.

## A Proof supplements

### A.1 The zero-zeta computation in one line

For completeness,

$$\sum_{n \geq 1} \frac{\nu_2(n)}{n^s} = \sum_{k \geq 1} \sum_{2^k | n} \frac{1}{n^s} = \zeta(s) \sum_{k \geq 1} 2^{-ks} = \frac{\zeta(s)}{2^s - 1}, \quad (125)$$

$$\sum_{n \geq 1} \frac{1 + \nu_2(n)}{n^s} = \frac{\zeta(s)}{1 - 2^{-s}}. \quad (126)$$

Dilation by  $M$  multiplies a Dirichlet series by  $M^{-s}$ , giving (59).

### A.2 Expansion at $s = 0$

Let  $a = \log_2 M$  and  $h = \log M = a \log 2$ . Then

$$\frac{1 - M^{-s}}{1 - 2^{-s}} = a \left( 1 + \frac{\log 2 - \log M}{2} s + \mathcal{O}(s^2) \right). \quad (127)$$

Multiplying by  $\zeta(s) = -\frac{1}{2} - \frac{1}{2} \log(2\pi)s + \mathcal{O}(s^2)$  gives (64)–(65).

### A.3 The finite-cascade coefficient recurrence

If  $C_{M,N}(x) = \sum_n c_{M,N}(n)x^n$ , then

$$c_{M,N+1}(n) = \sum_{j=0}^{M-1} c_{M,N}(n - j2^N). \quad (128)$$

This is the exact update used in `experiments.py`. It avoids floating-point convolution and makes the checks of [theorem 9.1](#) fully integral.

## B Corpus coverage matrix

| Corpus component                     | Existing coverage used                                                                                                          | Duplication avoided in this report                                                                                                                  |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Fabius/Rvachev exposition    | Fourier sinc product, random convolution, exact integer zero orders, Fabius/up normalization, moments and arithmetic background | No rederivation of the main endpoint asymptotic hierarchy, dyadic-value theory, or canonical primary exposition                                     |
| Thue–Morse atlas and frontier volume | Lacunary product $T(x)$ , Prouhet identities, Mahler equations, finite-block calculus, Mellin/Dirichlet machinery               | The new result is the positive digit quotient $A_M = T(x^M)/T(x)$ and its divisor interpretation, not another atlas of standard Thue–Morse formulas |
| Exponents and $q$ -series volume     | Bell–Bernoulli expansions, $q$ -binomial filters, general extrapolation machinery                                               | Only the new integer parameter $q = M^{-2}$ and divisor approximants are developed; general $q$ identities are cited as infrastructure              |
| Inverse and sampling volume          | Inverse endpoint expansions, Lambert $W_{-1}$ , dyadic sampling, comb formulas                                                  | The uniform quantile bracket and divisor torus law are distinct global statements; the existing all-orders inverse expansion is not duplicated      |
| Representation frontiers             | Legendre/Lagrange expansions, spline representations, multiresolution, polyphase structure                                      | The report gives a new divisor-indexed triangular transfer, not a replacement derivation of the existing representation loops                       |

| Corpus component      | Existing coverage used                                                      | Duplication avoided in this report                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontier compilations | Spectral zeta, zero filtration, transport metrics, general atomic functions | The new zeta belongs to the quotient zero divisor and carries the Euler factor $1 - M^{-s}$ ; odd singular approximants and complete scale classification were not found |
| Frozen archive        | Historical versions and provenance hashes                                   | Used only to prevent an absorbed result from being mistaken as absent; repeated archived text is not recounted as new coverage                                           |

## C Archive contents and file map

The repository package contains:

| Path                                                       | Purpose                                                      |
|------------------------------------------------------------|--------------------------------------------------------------|
| Fabius_Rvachev_Reciprocal_Integer_Convolution_Divisors.tex | complete LaTeX source                                        |
| Fabius_Rvachev_Reciprocal_Integer_Convolution_Divisors.pdf | compiled report                                              |
| experiments.py                                             | commented reproducibility script                             |
| figures/*.png                                              | four figures used in the PDF                                 |
| data/cascade_summary.csv                                   | finite-cascade maxima and $L^2$ energies                     |
| data/zero_multiplicities.csv                               | exact zero orders for sample $M$                             |
| data/odd_fourier_subsequence.csv                           | high-precision nondecaying Fourier values                    |
| data/stern_hyperbinary_check.csv                           | exact Stern coefficient comparison                           |
| data/endpoint_profiles.csv                                 | plotted endpoint phase data                                  |
| data/experiment_summary.txt                                | exact-check summary                                          |
| README.txt                                                 | reproduction and compilation instructions                    |
| Git history                                                | recoverable former package-checksum receipts; no live ledger |

## References

- [1] V. Reshetnikov, *Fabius Function and Rvachev Up*, living LaTeX exposition in the Analysis/FabiusFunction repository, [https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/Fabius\\_Function\\_and\\_Rvachev\\_Up](https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up), accessed 30 August 2026.
- [2] V. Reshetnikov, *Semi-formalized research frontiers and draft manifest*, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers>, accessed 30 August 2026.
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